Meghan Hsin-Ru Lee

Contact Information

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Education

University of California, Santa Barbara, Santa Barbara, CA

Ph.D., Mathematics September 2025 -

Wake Forest University, Winston-Salem, NC

M.S., Mathematics July 2023 - May 2025

- Master's Thesis: *Isolated j-invariants arising from the modular curve $X_0(n)$*
- Advisor: Abbey Bourdon

Occidental College, Los Angeles, CA

B.A., Mathematics (Cum Laude) August 2019 - May 2023

- Math Department Honors
- Distinction Pass on all 5 lower-division comprehensive exams (Calculus I-III, Linear Algebra, Discrete Math)
- Honors Senior Thesis: *The fundamental group and Van Kampen Theorem*
- Advisor: Don Lawrence

Awards

UC Santa Barbara *Eugene Cota-Robles Fellowship* September 2025 - June 2031
Wake Forest *Summer Graduate Research Fellowship* June 2024 - July 2024
Wake Forest *Pi Mu Epsilon Honors Society* April 2024
Occidental College *Benedict Freedman Mathematical Promise Prize* April 2023
Awarded to a junior or senior Occidental student who has demonstrated exceptional mathematical promise through original research in a mathematical science.
Occidental College *Honors Scholarship* August 2019 - May 2023

Papers

[**4**] *The exceptional set in Cassels's theorem on small cyclotomic integers* (with Jitendra Bajpai, Srijan Das, Kiran Kedlaya, Nam Le, Antoine Leudière, Jorge Mello). In preparation.

[**3**] *Isolated j-invariants arising from the modular curve $X_0(n)$*, preprint available at arxiv.org:2506.19560.

[**2**] *Complementation of subquandles* (with Kieran Amsberry, August Bergquist, Thomas Horstkamp, David Yetter). Involve, a Journal of Mathematics **18-3** (2025), 417-435.

[**1**] *The fundamental group and Van Kampen Theorem*, Occidental College honors thesis, 2023.

Conference & Colloquium Presentations

Isolated j-invariants arising from the modular curve $X_0(n)$, Regensburg GAP Days, Universität Regensburg. July 2025.

An algorithm for isolated j-invariants arising from $X_0(n)$, LMFDB, Computation, and Number Theory (LuCaNT), ICERM. July 2025.

Master's thesis defense: Isolated j-invariants arising from the modular curve $X_0(n)$, Wake Forest University, April 2025.

An algorithm for isolated points on $X_0(n)$, Contributed Paper Session on Number Theory and Field Theory, Joint Math Meetings (JMM). January 2025.

An algorithm for isolated points on $X_0(n)$, Southern California Number Theory Day 2024 Lightning Talks, University of California, Irvine. October 2024.

An algorithm for isolated points on $X_0(n)$, Palmetto Number Theory Series (PANTS) XXXVIII, Wake Forest University. September 2024.

Quandle constructions and examples, Benedict Freedman Award Reception, Occidental College. May 2023.

Senior honors thesis talk: The fundamental group and Van Kampen Theorem, Math department colloquium, Occidental College. March 2023.

Complementation of subquandles: An introduction to undergraduate math research, Math department colloquium, Occidental College. February 2023.

Complementation of subquandles, 25th Nebraska Conference for Undergraduate Women in Mathematics, University of Nebraska-Lincoln. January 2023.

Complementation of subquandles, Pi Mu Epsilon Undergraduate Student Poster Session, Joint Math Meetings (JMM). January 2023.

Subquandle lattices, REU Closing Conference, Kansas State University. August 2022.

Scholarly Experiences

Rethinking Number Theory 6, American Institute of Mathematics — July 2025
Workshop for building a collaborative number theory research community. Participated in a project computing the exceptional set in Cassel's theorem on classifying small cyclotomic integers, advised by Kiran Kedlaya

Introduction to the Theory of Algebraic Curves, Simons-Laufer Mathematical Sciences Institute (formerly MSRI) — July 2024
Graduate summer school covering approaches to studying algebraic curves, including Brill-Noether theory, moduli spaces of stable curves, and intersection theory

Research Training Program, Wake Forest University — January - May 2024
Developed and led an introductory research project for 2 undergraduates on tilings

Independent Study: Geometric Group Theory, Occidental College — January - May 2023
Selections from Clara Löh's *Geometric Group Theory, an introduction*, advised by Ramin Naimi and Timothy Rainone

Independent Study: Algebraic Topology, Occidental College — January - May 2023
Selections from Munkres's *Topology* and Hatcher's *Algebraic Topology*, advised by Don Lawrence

Undergraduate Math REU, Kansas State University — June 2022 - August 2022
Research project studying quandles via orbit decompositions and subquandle lattices, advised by David Yetter

Practicum for Undergraduates in Number Theory, Institute for Pure and Applied Math
October 2021
Undergraduate problem session series on modern research directions in number theory

Polymath Jr. REU June 2021 - August 2021
Readings from *The Knot Book*, followed by selected research papers extending classical knot theory to knotted surfaces in four dimensions, advised by Alexander Zupan

Teaching

University of California, Santa Barbara, Santa Barbara, CA Fall 2025 -
Teaching Assistant
- Math 34A: Calculus for Social and Life Sciences (Fall 2025)

Wake Forest University, Winston-Salem, NC Fall 2023 - Spring 2025
Teaching Assistant
- Math 345: Elementary Number Theory (Spring 2025)
- Math 111: Calculus with Analytic Geometry I (Spring 2025)
- Math 121: Linear Algebra (Fall 2024)
- Math 106: Calculus Foundations (Fall 2024)
- Math 105L: Precalculus Lab (Fall 2024)
- Math 321: Modern Algebra I (Spring 2024)
- Math 106: Calculus Foundations (Fall 2023)

Recitation Leader
- Math 111: Calculus with Analytic Geometry I (Spring 2025, Spring 2024, Fall 2023)

Math and Stats Center Tutor
Precalculus, differential and integral calculus, linear algebra, discrete math, undergraduate & graduate algebra, graduate topology

Occidental College, Los Angeles, CA Spring 2022 - Spring 2023
Grader
- Math 110: Calculus I (Spring 2023)
- Math 310: Real Analysis (Fall 2022)
- Math 210: Discrete Math (Spring 2022)

Service

Association for Women in Mathematics
AWM Newsletter Student Column Editor February 2025 - February 2027

Association for Women in Mathematics, Wake Forest University chapter
Graduate Student President August 2024 - May 2025
Summer Research Panel October 2024

Research Training Program, Wake Forest University
Graduate Student Project Leader January 2024 - May 2024

Integration Bee, Wake Forest University
Grader March 2024

MATHCOUNTS competition, Winston-Salem, NC chapter
Grader February 2024

Prison Mathematics Project
Mentor January 2023 - October 2023

Additional Experience

The Occidental Newspaper, Occidental College
Senior Editor May 2022 - May 2023
Editor-in-Chief December 2021 - May 2022
Managing Editor May 2021 - December 2021

News, Arts & Culture Section Editor May 2020 - May 2021
Writer, Photographer, Communications Director August 2019 - May 2023

International Model United Nations Association
Director of Conference Operations April 2020 - March 2021
Administrative Assistant Director August 2019 - March 2020

Student Leadership & Community Engagement, Occidental College
Orientation Team Leader July - August 2020

Neighborhood Partnership Program, Occidental College
College Applications Mentor August 2019 - January 2020

Technology literacy for senior citizens, Cerritos Senior Center, CA
Weekly Workshop Leader September 2017 - June 2019

COMPUTER SKILLS Magma, SageMath, Java, Python, Rust, Mathematica

Experience with NumPy and SnapPy Python libraries.